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In [1]: import pandas as pd
```

```
In [2]: # TASK 1: Convert Date Strings to Datetime
dates = pd.Series([
    "15-01-2026",
    "28-02-2026",
    "10-04-2026",
    "25-06-2026",
    "05-09-2026"])

dates = pd.to_datetime(dates, format="%d-%m-%Y")

print("TASK 1 - Converted Dates")
print(dates)
```

TASK 1 - Converted Dates

```
0    2026-01-15
1    2026-02-28
2    2026-04-10
3    2026-06-25
4    2026-09-05
dtype: datetime64[us]
```

```
In [3]: # TASK 2: Extract Year, Month and Day
movies = pd.DataFrame({"movie": ["Inception", "Avatar", "Titanic", "Interstellar", "Joker"],
    "release_date": ["16-07-2010", "18-12-2009", "19-12-1997", "07-11-2014", "02-10-2019"]})
```

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In [4]: movies["release_date"] = pd.to_datetime(movies["release_date"], format="%d-%m-%Y")
movies["year"] = movies["release_date"].dt.year
movies["month"] = movies["release_date"].dt.month
movies["day"] = movies["release_date"].dt.day

print("\nTASK 2 - Movie Release Date Details")
print(movies)
```

TASK 2 - Movie Release Date Details

	movie	release_date	year	month	day
0	Inception	2010-07-16	2010	7	16
1	Avatar	2009-12-18	2009	12	18
2	Titanic	1997-12-19	1997	12	19
3	Interstellar	2014-11-07	2014	11	7
4	Joker	2019-10-02	2019	10	2

```
In [8]: # TASK 3: Date Range for Next 7 Days
today = pd.Timestamp.today()

dates = pd.date_range(start=today, periods=7)

print("Next 7 Days:")
print(dates)
```

Next 7 Days:

```
DatetimeIndex(['2026-09-07 17:07:19.497529', '2026-09-08 17:07:19.497529',
    '2026-09-09 17:07:19.497529', '2026-09-10 17:07:19.497529',
    '2026-09-11 17:07:19.497529', '2026-09-12 17:07:19.497529',
    '2026-09-13 17:07:19.497529'],
    dtype='datetime64[us]', freq='D')
```

In [9]: *# TASK 4: Zomato Gold Subscription Tenure*

```
zomato_gold = pd.DataFrame({"user_id": ["U001", "U002", "U003", "U004", "U005"],  
"start_date": ["01-01-2026", "15-02-2026", "10-03-2026", "20-05-2026", "01-08-2026"],  
"tenure_days": [249, 204, 181, 110, 37]})  
zomato_gold["start_date"] = pd.to_datetime(zomato_gold["start_date"], dayfirst=True)  
zomato_gold["tenure_days"] = (pd.Timestamp.today() - zomato_gold["start_date"]).days  
  
print(zomato_gold)
```

	user_id	start_date	tenure_days
0	U001	2026-01-01	249
1	U002	2026-02-15	204
2	U003	2026-03-10	181
3	U004	2026-05-20	110
4	U005	2026-08-01	37

In []: